

LCS

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0.2 Longest Common Subsequence (LCS) function implementation

```
[1]: def lcs(s1, s2):  
    m, n = len(s1), len(s2)  
  
    # Create 2D table dp  
    dp = [[0] * (n + 1) for _ in range(m + 1)]  
  
    # Fill the table  
    for i in range(1, m + 1):  
        for j in range(1, n + 1):  
            if s1[i - 1] == s2[j - 1]:  
                dp[i][j] = dp[i - 1][j - 1] + 1  
            else:  
                dp[i][j] = max(dp[i - 1][j], dp[i][j - 1])  
  
    return dp[m][n]
```

0.3 lets introduce test samples

```
[2]: #short string sample
s1 = "ABCD"
s2 = "ACBD"
print("LCS length (short):", lcs(s1, s2))
```

LCS length (short): 3

medium string samples

```
[3]: s1 = "AGGTABCD"
s2 = "GXTXAYBD"
print("LCS length (medium):", lcs(s1, s2))
```

LCS length (medium): 5

longer string samples

```
[4]: s1 = "ABCBADABXYZ"
s2 = "BDCABAXYZZ"
print("LCS length (long):", lcs(s1, s2))
```

LCS length (long): 7